

IEM KOLKATA
Linear Algebra
Matrices & Determinants

Q1) What is the number of multiplications required to find $A^{4 \times 5} \times B^{5 \times 6}$?

Q2) If $A = [a_{ij}]^{m \times n}$ where $a_{ij} = (i + j) \forall i, j$. Find the sum of elements of A .

Q3) Find the number of different matrices with 4 elements whose elements can be either 0,1,2 or 3.

Q4) What is the transpose of a matrix A?

$$A = \begin{bmatrix} a & b & c \\ d & e & f \\ g & h & i \end{bmatrix}$$

(a) $\begin{bmatrix} a & b & c \\ d & e & f \\ g & h & i \end{bmatrix}$ (b) $\begin{bmatrix} a & d & g \\ b & e & h \\ c & f & i \end{bmatrix}$ (c) $\begin{bmatrix} a & c & b \\ g & i & h \\ d & f & e \end{bmatrix}$ (d) $\begin{bmatrix} a & e & i \\ b & f & g \\ c & h & d \end{bmatrix}$

Q5) What is the determinant of a matrix used in linear algebra?

- (A) To determine if a matrix is symmetric.
- (B) To determine whether a matrix is invertible.
- (C) To assess the linear independence of vectors.
- (D) To calculate the rank of the matrix.

Q6) Calculate the determinant of the matrix:

$$M = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{bmatrix}$$

Q7) If B is a skew-symmetric matrix, what type of matrix is B^{100} ?

- (a) Skew-symmetric matrix
- (b) Symmetric matrix
- (c) Identity matrix
- (d) Diagonal matrix

Q8) If A and B are 2×2 matrices, which of the following is true?

- (a) $(A + B)(A - B) = A^2 - B^2$
- (b) $(A - B)(A - B) = A^2 + B^2 - 2AB$
- (c) $(A + B)(A + B) = A^2 + B^2 + 2AB$
- (d) $(A - B)(A + B) = A^2 + AB - BA - B^2$

Q9) Let A and B be $n \times n$ matrices where A is invertible and B is singular. Which of the following statements is always true?

- (a) $A + B$ is invertible.
- (b) AB is singular.
- (c) B^2 is singular.
- (d) $A^{-1}B$ is invertible.

Q10) The determinant of the matrix:

$$\begin{bmatrix} 6 & -8 & 1 & 1 \\ 0 & 2 & 4 & 6 \\ 0 & 0 & 4 & 8 \\ 0 & 0 & 0 & -1 \end{bmatrix}$$

is:

- (a) 11 (b) -48 (c) 0 (d) -24

Q11) Let A be a $m \times n$ matrix and B be a $n \times p$ matrix. Which of the following is always true regarding the rank of the product AB ?

- (a) $\text{Rank}(AB) \leq \text{Rank}(A)$
- (b) $\text{Rank}(AB) \geq \text{Rank}(B)$
- (c) $\text{Rank}(AB) \leq \min(\text{Rank}(A), \text{Rank}(B))$

(d) $\text{Rank}(AB) = \max(\text{Rank}(A), \text{Rank}(B))$

Q12) Let A and B be two $n \times n$ matrices over real numbers. Let $\text{rank}(M)$ and $\det(M)$ denote the rank and determinant of a matrix M, respectively. Consider the following statements.

I. $\text{rank}(AB) = \text{rank}(A) \times \text{rank}(B)$

II. $\det(AB) = \det(A) \times \det(B)$

III. $\text{rank}(A+B) \leq \text{rank}(A) + \text{rank}(B)$

IV. $\det(A+B) \leq \det(A) + \det(B)$

Which of the above statements are TRUE?

(a) I and II only

(b) I and IV only

(c) II and III only

(d) III and IV only

Q13) $A = \begin{bmatrix} 1 & 2 & 3 \\ 2 & 4 & 6 \\ 3 & 6 & 9 \end{bmatrix}$. Find $\text{Rank}(A)$.

Q14) $A = \begin{bmatrix} 1 & 1 & 2 & 4 \\ 1 & 2 & 3 & 4 \\ 2 & 4 & 6 & 8 \\ 3 & 6 & 9 & 12 \end{bmatrix}$. Find $\text{Rank}(A)$.

Q15) $A = [a_{ij}]^{m \times n}$ where $a_{ij} = 5 \forall i, j$. Find $\text{Rank}(A)$.

Q16) $A = \begin{bmatrix} 4 & 2 & 1 & 3 \\ 6 & 3 & 4 & 7 \\ 2 & 1 & 0 & 1 \end{bmatrix}$. Find $\text{Rank}(A)$.